hub, expanders, 3.3 V

cad_sch

Port 1

USB-C 1: switch, current sense, ESD, RESET/BOOT

File: port1.kicad_sch

Port 2

USB-C 2: switch, current sense, ESD, RESET/BOOT

File: port2.kicad_sch

Port 3

USB-C 3: switch, current sense, ESD, RESET/BOOT

File: port3.kicad_sch

Port 4

USB-C 4: switch, current sense, ESD, RESET/BOOT

File: port4.kicad_sch

INT, PORT1_BOOT_IO, PORT1_DN/DP/EN, PORT1_FLT_N, PORT1_RST_IO, SCL, SDA

INT, SCL, SDA

INT, SCL, SDA

INT, SCL, SDA

INT, PORT2_BOOT_IO, PORT2_DN/DP/EN, PORT2_FLT_N, PORT2_RST_IO, SCL, SDA

INT, SCL, SDA

INT, SCL, SDA

INT, PORT3_BOOT_IO, PORT3_DN/DP/EN, PORT3_FLT_N, PORT3_RST_IO, SCL, SDA

INT, SCL, SDA

INT, PORT4_BOOT_IO, PORT4_DN/DP/EN, PORT4_FLT_N, PORT4_RST_IO, SCL, SDA

DATA ×4 v1.0 — USB2514B hub, four USB-C (sub-slots A1.1-A1.4) each with a TPS2553-1 switch, an INA226 and ESD, two TCA6408A for EN/FAULT/RESET/BOOT, an AP2112K 3.3 V; the PH 8 lead from the bay node's J7 into J1 at the rear end of the right edge. 100 × 37 mm.
Lead pinout here: 1 INT · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 +5V (the node's J7 mirrored: one 'Same direction' PH 8 cable).
I²C: TCA6408A 0x20 (ports 1-2) / 0x21 (ports 3-4), INA226 0x41 / 0x43 / 0x44 / 0x45 (A1.1-A1.4); the node's INA226 stays alone at 0x40.

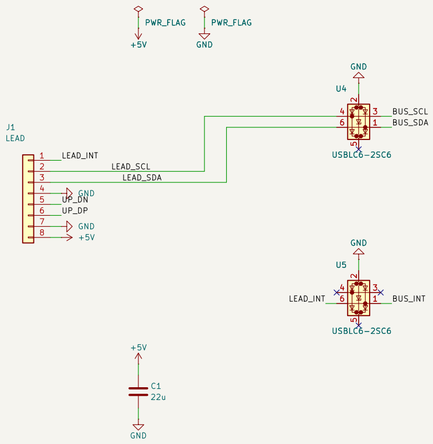
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi
hw-testing-rack

Sheet: /
File: data-port-x4.kicad_sch

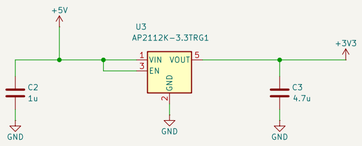
Title: DATA ×4 — Overview

Size: A3	Date:	**Rev: 1.0**
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Lead from the bay node's J7 (JST PH 8, side entry): SDA, SCL and INT through USBLC6s

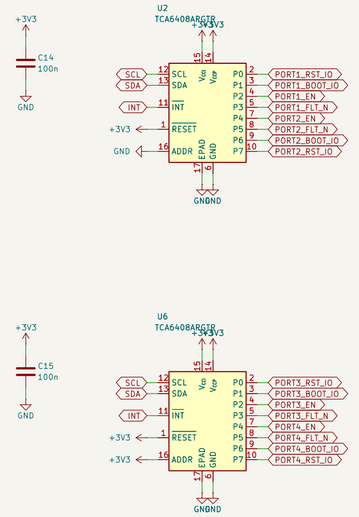


3.3 V for the hub, the expanders and the INA226s (AP2112K, ≈ 50 mV dropout at 0.1 A)



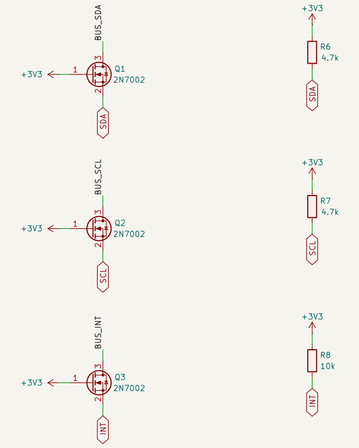
USB2514B hub on its straps (CFG_SEL 00: self-powered, one transaction translator per port), 24 MHz crystal

Two TCA6408A expanders, one per pair of ports: EN, FAULT_N, RESET, BOOT



Mounting: PT 2.6 × 8 into the drawer's floor bosses

I²C isolation: open with DUT_5V off, so the node's pull-ups cannot feed this board



Hub on its straps: CFG_SEL[1:0] = 00 and NON_REM[1:0] = 00, every strap pin on a 100k to GND (NON_REM1 shares CFG_SEL0's): self-powered, one TT per port; OCS_N left on their pull-ups (the expander reads each FAULT). PRTPWR unused: the switches follow the expander. VBUS_DET on 3.3 V (host always attached).
Expander pins are inputs at power-up: EN pulled down (every port off until the firmware enables it within the bay's budget). RESET/BOOT pulled up (released).
INT (J1.1 → the node's IO4): both expanders' INT and the four INA226 ALERTs, open drain, wired-OR, through Q3.
I²C and INT: the node's pull-ups on BUS_*, 4.7k / 4.7k / 10k here: with DUT_5V off the chips here lose their registers: the firmware sets them up again after each DUT_5V on, and leaves 0x20/0x21/0x41-0x45 alone while off.
Lead: the node's J7 mirrored, one 'Same direction' PH 8 cable. The upstream pair has no ESD here: the lead stays plugged inside the drawer, the node protects its own end.

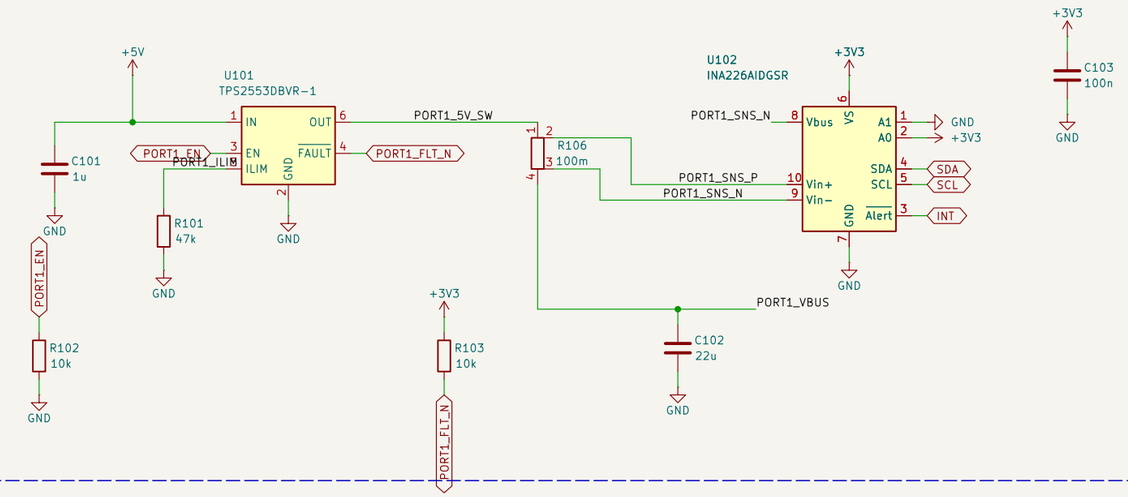
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A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi
hw-testing-rack

Sheet: /Hub/
File: hub.kicad_sch

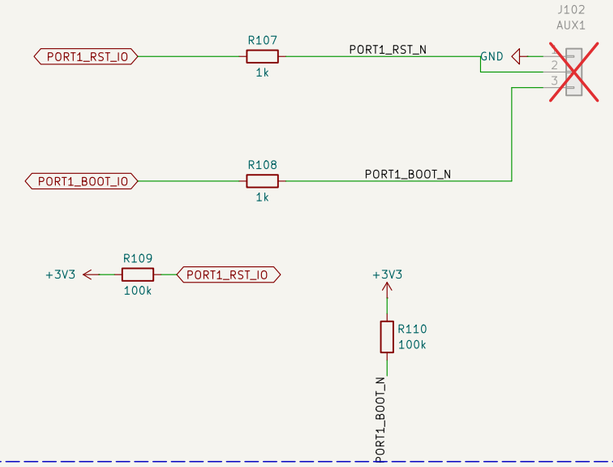
Title: DATA ×4 — Lead, hub, expanders, 3.3 V

Size: A2	Date:	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 2/6

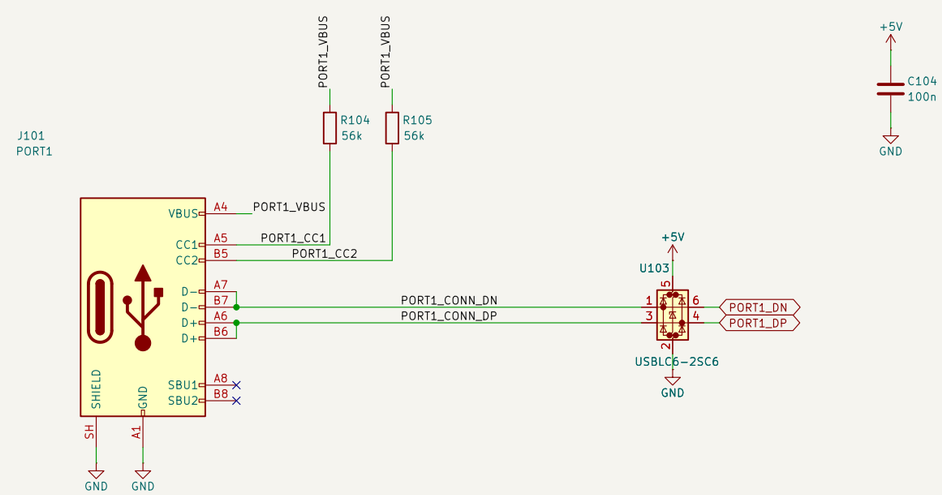
Port 1 power: +5V → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt



RESET / BOOT 1: through 1k to a 1 × 3 2.54 mm header (not fitted)



USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 1 = sub-slot A1.1 (the bay's position,
then the port), hub port 1, busid 1-1.1.
INA226 at 0x41; expander U2 (0x20):
P0 RST, P1 BOOT, P2 EN, P3 FLT_N.
ILIM 47k: 0.50-0.62 A (TI's IOS equations).
The −1 limits for 7.5 ms, then latches off
until EN toggles; FAULT_N flags it.
EN pulled down: the port is off until the
firmware enables it within the bay's budget.
RESET/BOOT: output register 0; the pin as an
output pulls low, as an input the 100k holds it high.
Pigtail: 1 GND, 2 RESET_N, 3 BOOT_N (Dupont
female jumpers to the board's own pins).

Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi

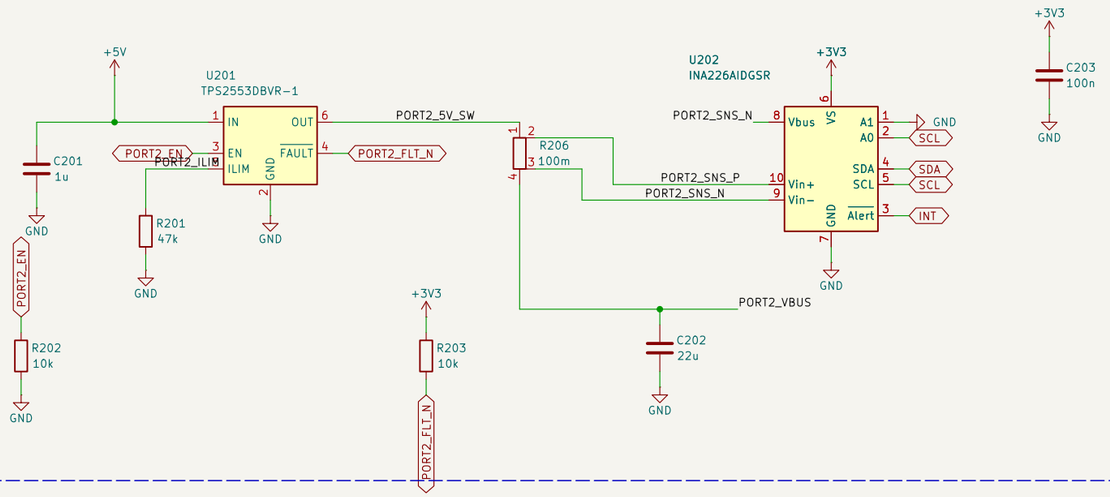
hw-testing-rack

Sheet: /Port 1/
File: port1.kicad_sch

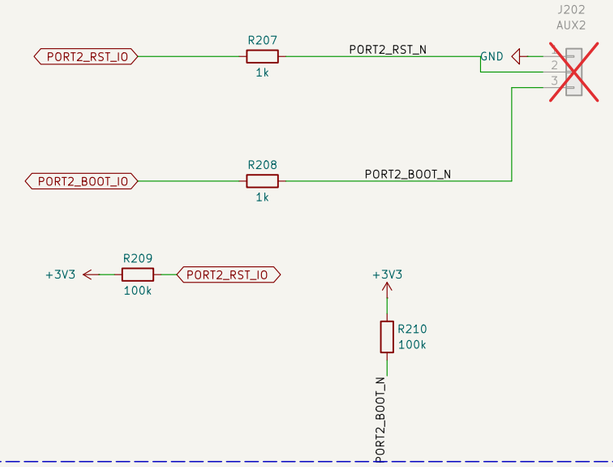
Title: DATA ×4 — USB-C 1: switch, current sense, ESD, RESET/BOOT

Size: A3	Date:	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 3/6

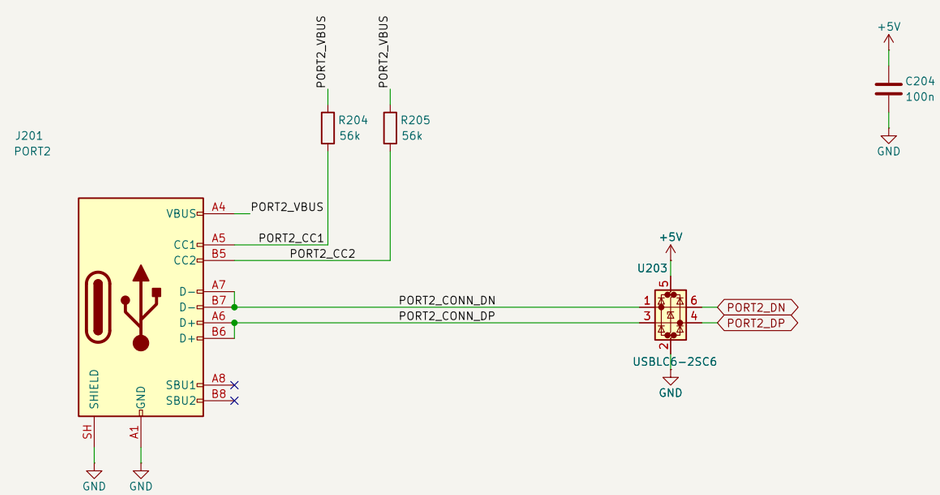
Port 2 power: +5V → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt



RESET / BOOT 2: through 1k to a 1 × 3 2.54 mm header (not fitted)



USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 2 = sub-slot A1.2 (the bay's position,
then the port), hub port 2, busid 1-1.2.
INA226 at 0x43; expander U2 (0x20):
P4 EN, P5 FLT_N, P6 BOOT, P7 RST.
ILIM 47k: 0.50-0.62 A (TI's IOS equations).
The -1 limits for 7.5 ms, then latches off
until EN toggles; FAULT_N flags it.
EN pulled down: the port is off until the
firmware enables it within the bay's budget.
RESET/BOOT: output register 0; the pin as an
output pulls low, as an input the 100k holds it high.
Pigtail: 1 GND, 2 RESET_N, 3 BOOT_N (Dupont
female jumpers to the board's own pins).

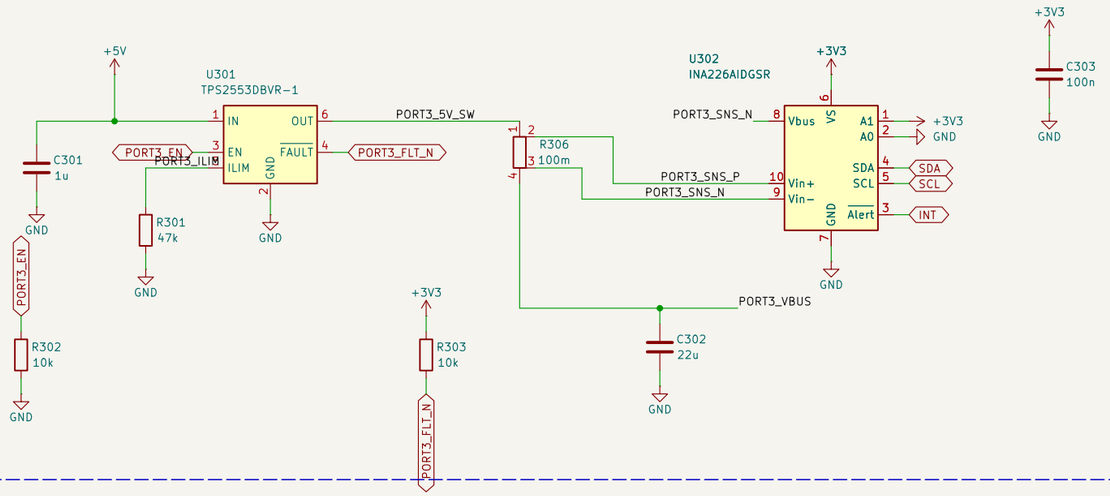
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi
hw-testing-rack

Sheet: /Port 2/
File: port2.kicad_sch

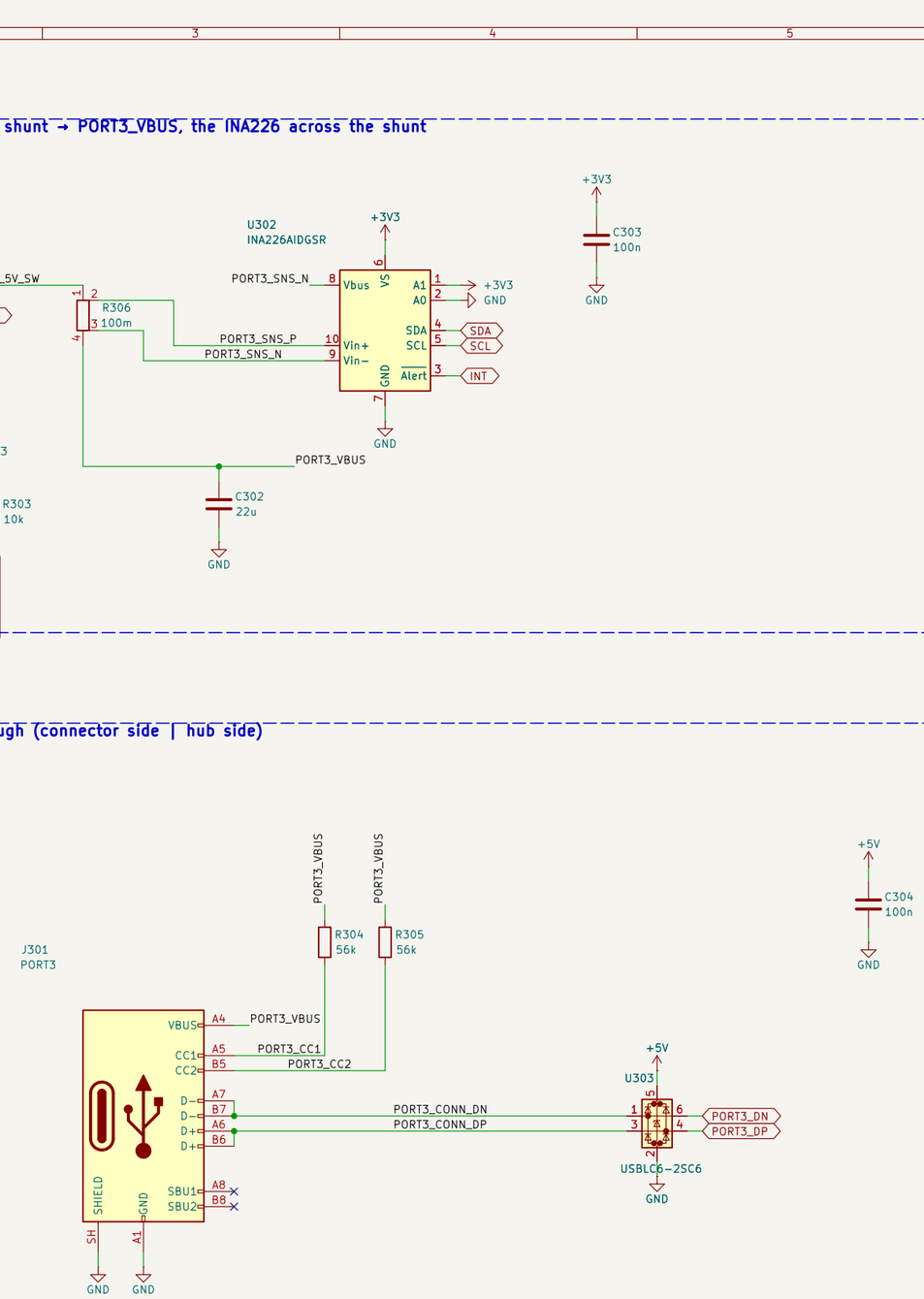
Title: DATA ×4 — USB-C 2: switch, current sense, ESD, RESET/BOOT

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/6

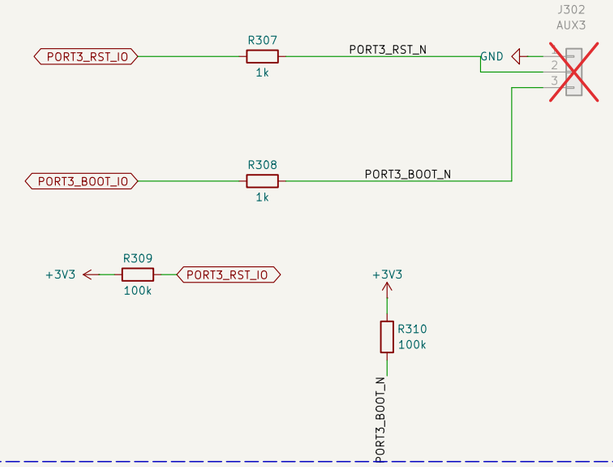
Port 3 power: +5V → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt



USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



RESET / BOOT 3: through 1k to a 1 × 3 2.54 mm header (not fitted)



Port 3 = sub-slot A1.3 (the bay's position,
then the port), hub port 3, busid 1-1.3.
INA226 at 0x44; expander U6 (0x21):
P0 RST, P1 BOOT, P2 EN, P3 FLT_N.
ILIM 47k: 0.50-0.62 A (TI's IOS equations).
The −1 limits for 7.5 ms, then latches off
until EN toggles; FAULT_N flags it.
EN pulled down: the port is off until the
firmware enables it within the bay's budget.
RESET/BOOT: output register 0; the pin as an
output pulls low, as an input the 100k holds it high.
Pigtail: 1 GND, 2 RESET_N, 3 BOOT_N (Dupont
female jumpers to the board's own pins).

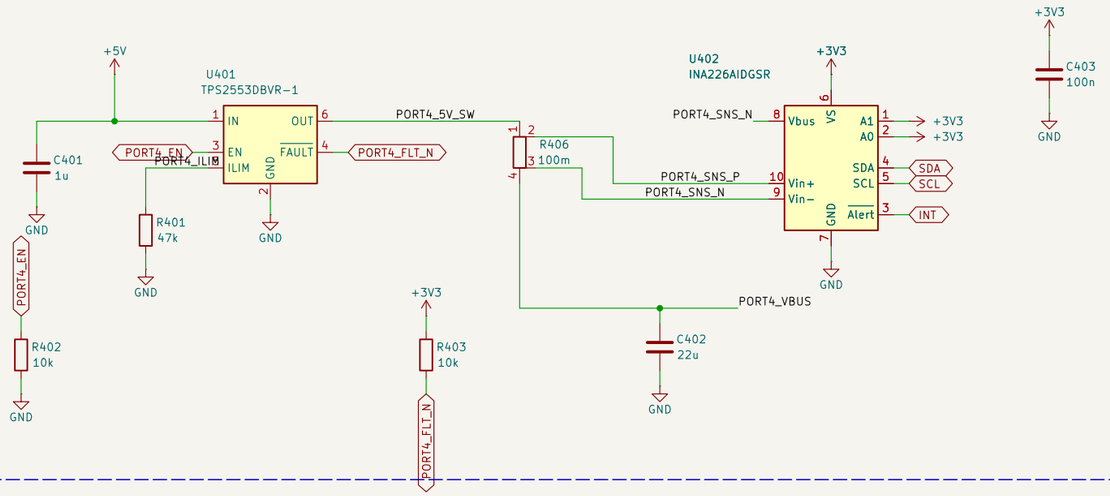
Generated from electronics/data-port-x4/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi
hw-testing-rack

Sheet: /Port 3/
File: port3.kicad_sch

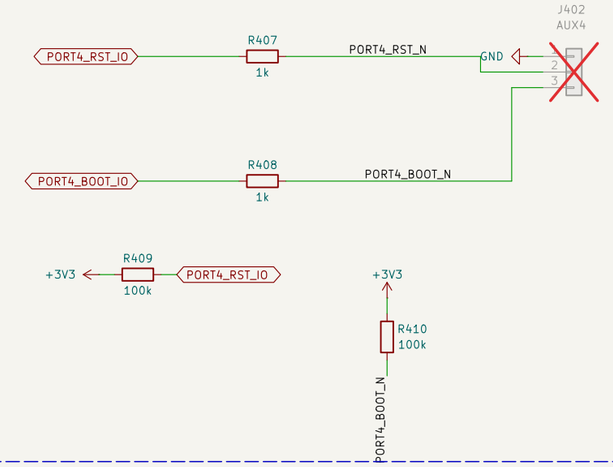
Title: DATA ×4 — USB-C 3: switch, current sense, ESD, RESET/BOOT

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 5/6

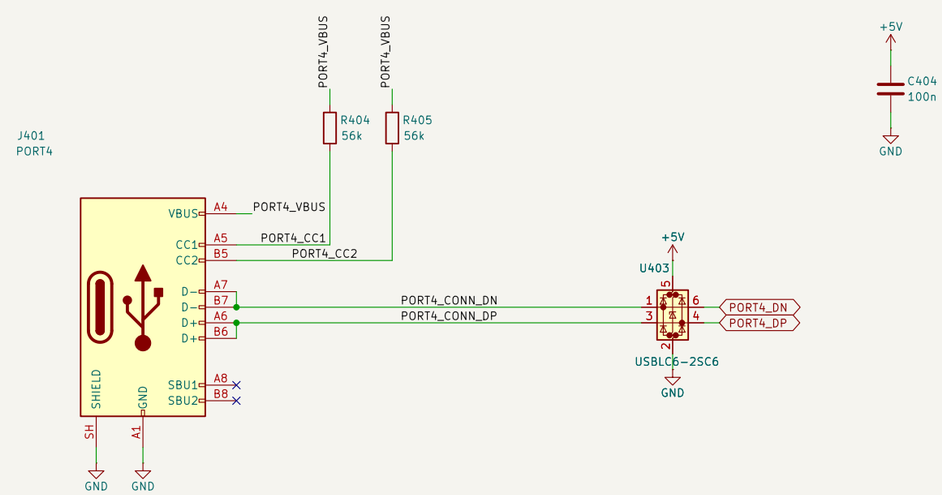
Port 4 power: +5V → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt



RESET / BOOT 4: through 1k to a 1 × 3 2.54 mm header (not fitted)



USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Port 4 = sub-slot A1.4 (the bay's position,
then the port), hub port 4, busid 1-1.4.
INA226 at 0x45; expander U6 (0x21):
P4 EN, P5 FLT_N, P6 BOOT, P7 RST.
ILIM 47k: 0.50-0.62 A (TI's IOS equations).
The −1 limits for 7.5 ms, then latches off
until EN toggles; FAULT_N flags it.
EN pulled down: the port is off until the
firmware enables it within the bay's budget.
RESET/BOOT: output register 0; the pin as an
output pulls low, as an input the 100k holds it high.
Pigtail: 1 GND, 2 RESET_N, 3 BOOT_N (Dupont
female jumpers to the board's own pins).

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A converted bay's drawer board: four USB-C behind a hub, switched and measured per port (hw-testi
hw-testing-rack

Sheet: /Port 4/
File: port4.kicad_sch

Title: DATA ×4 — USB-C 4: switch, current sense, ESD, RESET/BOOT

Size: A3	Date:	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 6/6